Appendix A

Ineterview with Onno, CRM Track

1. Can you describe your role in relation to the student research projects you have been involved in or supervised?

I was a student project lead within the Value Chain Hackers Lab. I was responsible for coordinating the project work, keeping contact with stakeholders, and making sure that the research activities were done as agreed. I did not supervise other students in a formal way, but I did take responsibility for planning, communication, and follow-up within the team.

2. At the start of projects, what guidance are students typically given about how they are expected to document their research work? Can you give an example?

At the start, they said we should put our work somewhere shared, so that was fine. I did not really think too much about it beyond that. I focused on doing the actual work and making progress. Things like notes from conversations or small observations, I mostly kept in my head or in my own files, because I felt I knew what was relevant. For me, the important part was the final outcome and whether the solution made sense, not writing everything down along the way.

3. Can you describe specific situations where documentation came up during project work with students? What happened in those moments?

Documentation usually came up when someone was missing context. For example, when a stakeholder asked why a certain choice was made, or when a supervisor joined later in the project. In those moments, we noticed that decisions were discussed verbally but not always written down, so we had to reconstruct the reasoning afterwards.

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to the original student team? Can you give an example?

Someone new could understand what we were working on, but it would probably take time to see how we got there. A lot of ideas developed through discussions and working sessions, and those steps are not always visible in the files. You see the result, but not everything that led up to it.

5. Which parts tend to be difficult for someone new to understand based only on what is written down or stored?

The link between the stakeholder conversations and the concepts might be hard to follow. We discussed many things as a team before shaping them into the final ideas, and not all of those discussions were captured. Without that background, some choices are probably unclear.

6. Can you think of an example of an important decision or change during a project, and describe how and where it was documented, if at all?

Yes. In our project, we changed the focus of the research after the first interviews. This was discussed in meetings and agreed with the stakeholder, but it was mainly documented implicitly in later versions of the research question.

7. If someone new wanted to understand how one important conclusion of a project was reached, what would they need to look at or read?

They would need to look at multiple things. The final report, the interview notes, and possibly emails or meeting notes. But these are often stored in different places, so it takes effort to connect them.

8. During projects, is it clear who is responsible for documenting what, if anyone? How is that responsibility typically communicated or decided?

We never really defined responsibility for documentation. Everyone kept track of their own work, but there was no shared approach for organizing it. Since our focus was mainly on developing the concepts, documentation was not something we structured as a team.

9. What tools, templates, or systems are students expected to use to document work, and where is everything actually stored in practice?

We were expected to use tools like Word and Google Docs and to store documents on something shared like GitHub or a Google Drive. In practice, it was more a collection of people’s personal notes and files on their own devices that were later brought together when needed, rather then a central place.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, influences how motivated students are to document their work well?

I would assume to offer clarity on the project and the process we undertook. Supervisors, future students, and stakeholders benefit from it. Students are more likely to document when they see the point in it, if it feels like an extra task instead of part of the submission process i'm not sure how likely they would be to do that.

11. Based on your experience, what tends to get lost most often over time across projects, for example context, decisions, data, analysis steps, or contacts?

Over time it becomes less clear how the ideas developed. You can still see the final concepts, but the earlier discussions and intermediate thinking are less visible. That makes it harder for someone else to understand how the project progressed.

12. How does documentation fit into project work from your perspective? Does it seem integrated into the work, or more like something added on top? Why?

At the start we were mainly focused on developing the concepts, so documentation was not something we discussed much as a team. It became more relevant once we had to organize the material, but there was no fixed way of doing it during the project.

13. From your perspective, what currently makes it difficult to achieve consistent documentation across projects, and what would need to be in place for research outputs to be considered trustworthy and usable over time?

What makes it difficult is unclear expectations and lack of structure. To improve this, there should likely be clearer guidelines and expectations and a single system where everything is stored. Also, documenting decisions during the project would help.

14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?

I think continuity between projects is important. Often, projects build on each other, but knowledge is not transferred well. Better documentation would help new teams start faster and avoid repeating the same mistakes.

Interview with Quina, Cacao Track

1. Can you describe your role in relation to the research project or projects you have been involved in?

I was the project lead for the Cocoa track. I helped coordinate the team's work, stayed in contact with stakeholders together with others, and supported the group in keeping the project on schedule

2. At the start of the project, what guidance were you given about how you were expected to document your research work? Can you give an example?

We were encouraged to keep our materials in shared spaces so the team could access them when needed. Beyond that, the approach developed naturally as we collaborated

3. Can you describe specific situations where documentation came up during the project work? What happened in those moments?

Doocumentation usually became relevant when we tried to bring our ideas together into something coherent. Putting things into writing helped align everyone's understanding.

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to you? Can you give an example?

The report is readable, but not very easy to reuse. Someone can see the conclusions, but not exactly how I got there. For example, interviews and sources are mentioned, but the steps in between are not always clear. Without talking to me, it would be difficult to continue the work.

5. Which parts would be difficult for someone new to understand based only on what is written down or stored?

The reasons behind decisions. Why I chose one direction and not another. Also assumptions made about the cocoa chain are not always written down.

6. Can you think of an example of an important decision or change during the project, and describe how and where it was documented, if at all?

The scope of the project changed during the process because of time and data limits. This decision was not clearly documented as a decision. It only shows in the final report that the focus is smaller, without further note.

7. If someone new wanted to understand how one important conclusion was reached, what would they need to look at or read?

They would need more than just the report. They would need notes, drafts, and maybe extra explanation.

8. During the project, was it clear who was responsible for documenting what, if anyone? How was that responsibility communicated or decided?

We treated documentation as a collective effort of necessity and task assignment. Everyone contributed in their own way as the project moved forward.

9. What tools, templates, or systems were used to document work, and where was everything stored in practice?

Mainly Word documents and google folders. There was no fixed template or system for documentation. Things were stored in different places, like drafts, notes, and documents, which made it a bit messy.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, motivates students to document their work well?

Documentation should make the research understandable for others. Especially for future students or supervisors. In practice, motivation mainly comes from assessment. If documentation is not clearly graded or required, it is easier to focus on finishing the project instead.

11. Based on your experience, what tends to get lost most often over time, for example context, assumptions, decisions, data, analysis steps, or contacts?

After some time, it is harder to see how the team's conversations shaped the direction we took. What stays visible is the final work, but not always how everyone's input contributed to it.

12. How did documentation fit into the project work overall? Did it feel integrated into the work, or more like something added on top? Why?

Our main focus was developing the work together, so writing usually happened after those discussions rather than during them. It kind of followed the project inherently,

13. If you were starting the project again, what would you personally do differently in how you document your work?

I would try to capture important discussions earlier, so it stays clearer how the team reached certain ideas.

14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?

Something I noticed is that when projects move from one team to another, small pieces of practical knowledge are not always carried forward. Having a simple way to pass that on would make it easier for the next group to continue without starting from zero

Interview with Kim,, the Bakery Track.

1. Can you describe your role in relation to the research project or projects you have been involved in?
I was the lead for the Bakery project. That meant keeping the team focused, making decisions when things were unclear, and making sure the final submission was completed.

2. At the start of the project, what guidance were you given about how you were expected to document your research work? Can you give an example?
The guidance was very general. We were told documentation is important and that we should use something like GitHub or a shared drive, but there was no clear explanation of what needed to be documented or what would actually be enough. So we mostly treated documentation as producing the final report.

3. Can you describe specific situations where documentation came up during the project work? What happened in those moments?
When reviewing earlier parts of the project, we sometimes noticed that certain choices were never clearly recorded. At the time everything felt obvious, but later it was harder to explain why we had moved in a particular direction.

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to the original team? Can you give an example?
The report can be understood, but continuing the work would be difficult without speaking to the team. Key steps behind the conclusions are not always visible, so someone new would likely need additional explanation before building on it

5. Which parts would be difficult for someone new to understand based only on what is written down or stored?
What tends to stay unclear is why certain directions were chosen along the way. Many of those decisions happened during team discussions and were not fully captured, which makes the project harder to interpret afterwards.

6. Can you think of an example of an important decision or change during the project, and describe how and where it was documented, if at all?
Partway through the project, we reduced the scope because of time and access constraints. That shift was agreed in conversations, but it was never clearly documented, so you only notice it when comparing earlier work with the final version.

7. If someone new wanted to understand how one important conclusion was reached, what would they need to look at or read?
To trace one conclusion back properly, you would need access to notes, draft material, and the earlier inputs that informed it. Since those were not kept together in one place, most people would probably rely on the team to explain it.

8. During the project, was it clear who was responsible for documenting what, if anyone? How was that responsibility communicated or decided?
No, that was not clear. People managed their own material, but no one coordinated how decisions or progress were recorded, so the structure depended largely on individual work and group sessions.

9. What tools, templates, or systems were used to document work, and where was everything stored in practice?
We used a mix of tools. Documents and slides, sometimes GitHub, sometimes shared drives, and also notes in messages or personal files. Where something was stored depended on who created it.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, motivates students to document their work well?
Documentation helps others understand what was done and continue the work later. Future students, supervisors, and stakeholders would likely benefit from that. When most attention goes to finishing the final output, documenting the process naturally becomes less of a priorit

11. Based on your experience, what tends to get lost most often over time, for example context, assumptions, decisions, data, analysis steps, or contacts?
Early context tends to disappear first, especially the discussions that influenced key decisions. Intermediate steps are often harder to trace later, and contact details sometimes remain in personal inboxes. What usually stays is the final version, without much insight into how it developed

12. How did documentation fit into the project work overall? Did it feel integrated into the work, or more like something added on top? Why?
It felt added on top. The priority was delivering the final output. Documentation was something you did afterwards or when someone asked for it. There was no workflow that naturally integrated documentation into daily work.

13. If you were starting the project again, what would you personally do differently in how you document your work?
I would keep it simple but consistent from the start. A clear expectation of how and what to do, including where to store it and making that part of the requirement for passing would allow that to become a part of the learning process. Maybe include everyones personal work and notes as well.

14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?

Yes i think relying on previous students for explanations is also difficult, so having a basic standard would make projects easier to understand without needing the students to give understanding to the project.

INterview with Kamile, Beer track.

1. Can you describe your role in relation to the research project or projects you have been involved in?
I worked on the Beer project as the lead researcher. My involvement focused on gathering information, speaking with people connected to the case, and helping shape our understanding of how things operated locally

2. At the start of the project, what guidance were you given about how you were expected to document your research work? Can you give an example?
At the beginning, the expectation was simply that our work should be stored somewhere shared so others could access it later. Beyond that, there was not much detail, so I mainly concentrated on understanding the case rather than thinking about how each step should be recorded

3. Can you describe specific situations where documentation came up during the project work? What happened in those moments?
The need to document became more noticeable once our project progressed. Earlier impressions sometimes changed as we learned more, and without notes it was harder to retrace how that shift had happened

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to the original team? Can you give an example?
The final report explains recycling practices in Romania, but a lot of the background sat in working files rather than in the submission itself. Many of our notes stayed on a shared Google Drive, so without looking at those, it would be harder for someone new to understand how our view developed or what shaped it.

5. Which parts would be difficult for someone new to understand based only on what is written down or stored?
The interpretation part. Why certain things were considered important and others less. Also the local context, which was understood during the research but not included in the final submission. Without being there or talking to us, it would be hard to fully define the context and process..

6. Can you think of an example of an important decision or change during the project, and describe how and where it was documented, if at all?

We decided at some point to focus more on bottle reuse and return systems, instead of recycling in general. This happened slowly, through discussions and learning more about the case. I do not think this was clearly documented as a decision. It just appeared in the work later.

7. If someone new wanted to understand how one important conclusion was reached, what would they need to look at or read?
They would need to see notes, early drafts, and maybe emails or messages where we discussed ideas. A lot of the work was developed through group talk and notes that were captured in shared files but not fully carried into the submission

8. During the project, was it clear who was responsible for documenting what, if anyone? How was that responsibility communicated or decided?
We did not spend much time defining who should document what. As the project moved forward, everyone kept track of their own findings in the way that suited them, so the material naturally spread across different files

9. What tools, templates, or systems were used to document work, and where was everything stored in practice?
As i mentioned most working material ended up on a shared Google Drive, where we stored notes, drafts, and reference items alongside the report itself. Since the final deliverable only reflected part of that material, someone relying solely on it would miss some of the background that informed our understanding.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, motivates students to document their work well?
I think documentation should help others understand the research later. Also for supervisors to see what was done. But it is not always clear why we document, so motivation is not very strong. Students focus more on finishing the project than on documenting every step.

11. Based on your experience, what tends to get lost most often over time, for example context, assumptions, decisions, data, analysis steps, or contacts?
The smaller details fade the fastest, especially the things you notice while being there. After some time, what remains is mostly the report, while those practical impressions are harder to recall.

12. How did documentation fit into the project work overall? Did it feel integrated into the work, or more like something added on top? Why?
It felt added on top. We worked on the research and then later tried to document it. Documentation was not something that guided the work step by step.

13. If you were starting the project again, what would you personally do differently in how you document your work?
I would write more notes during the process, even if they are informal. Especially about why we

thought something was important. And I would try to keep everything in one place, so it is easier to follow later.

14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?
I think in international projects, a lot of understanding comes from being close to the case. If this is not written down, the project loses depth. Documentation should help keep that understanding, not only the final conclusions.

Interview With Maxime , Lab Supervisor

1. Can you describe your role in relation to the student research projects you have been involved in or supervised?

I am involved in student research projects primarily as a program manager and researcher. Within Value Chain Hackers, my role is to ensure overall continuity and coherence across projects, across cohorts, and across years. This includes aligning student work with longer-term research agendas, external partnerships, and institutional requirements.

I also supervise and guide students during internships and applied research projects, with a focus on helping them translate exploratory work into outputs that can be communicated, assessed, and, where possible, reused.

2. At the start of projects, what guidance are students typically given about how they are expected to document their research work? Can you give an example?

At the start of projects, students are informed that documentation is a necessary part of applied research, both for academic accountability and for continuity across project cycles. This includes producing a clear final report, storing materials centrally, and making sources and methods explicit.

Documentation is framed as part of professional research practice, particularly because projects may feed into publications, funding accountability, or legal and institutional obligations.

3. Can you describe specific situations where documentation came up during project work with students? What happened in those moments?

Documentation often becomes particularly relevant at transition points. For example, when a project is handed over to a new group of students, or when results need to be reviewed for publication or external reporting.

In those moments, it becomes clear which parts of the work are sufficiently documented to stand on their own, and which parts rely heavily on verbal explanation or personal memory.

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to the original student team? Can you give an example?

The usability of final outputs varies. In many cases, reports provide a solid overview of the topic and the conclusions reached, which is valuable for orientation.

However, for someone who wants to build directly on the work, additional clarification is often needed. This is one of the reasons why structured documentation and publication-ready outputs are important: they reduce dependency on the original team.

5. Which parts tend to be difficult for someone new to understand based only on what is written down or stored?

What is often difficult to reconstruct are the intermediate steps. This includes how research questions evolved, how data limitations were handled, and how methodological choices were made under practical constraints.

These aspects are understandable within the educational context, but they do limit reuse if they are not made explicit.

6. Can you think of an example of an important decision or change during a project, and describe how and where it was documented, if at all?

A common example is a shift in scope once students encounter feasibility issues, such as limited access to data or stakeholders. These shifts are usually discussed with supervisors and reflected in the final framing of the project.

In some cases, the rationale for these changes is captured in reports or reflection sections, but not always in a way that is easily traceable for future users.

7. If someone new wanted to understand how one important conclusion of a project was reached, what would they need to look at or read?

They would need the final report, access to the datasets or sources used, and a clear description of the analytical approach. When these elements are present and well-structured, it becomes much easier to assess and reuse the work.

This is particularly important when research outputs are intended for publication or longer-term use.

8. During projects, is it clear who is responsible for documenting what, if anyone? How is that responsibility typically communicated or decided?

Responsibility for documentation is generally shared among students, with supervisors providing guidance and feedback. There is usually no single person formally assigned as documentation owner.

This works reasonably well for educational purposes, but it does create variability in how consistently documentation is handled across projects.

9. What tools, templates, or systems are students expected to use to document work, and where is everything actually stored in practice?

Students use a range of tools, including shared drives, reports, spreadsheets, and collaborative platforms. Templates are sometimes provided, particularly for reports or publications, to ensure a minimum level of consistency.

Storage is typically centralised, but the structure and completeness of project archives can differ from one project to another.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, influences how motivated students are to document their work well?

Documentation serves several purposes. It supports assessment, enables publication, meets legal and institutional requirements, and helps ensure continuity across student cohorts.

Students are generally more motivated to document their work when they see that it has a concrete purpose beyond grading, for example when it contributes to a publication, a follow-up project, or a real external deliverable.

11. Based on your experience, what tends to get lost most often over time across projects?

What tends to be lost is contextual knowledge: why certain compromises were made, how constraints shaped decisions, and what was learned informally during the process.

This is one of the challenges of working with student-led projects that have a limited timeframe.

12. How does documentation generally fit into project work from your perspective? Does it seem integrated into the work, or more like something added on top? Why?

Documentation is partially integrated, but it often becomes more prominent toward the end of projects. This reflects the reality of applied education, where students are balancing learning, experimentation, and deliverables within a fixed timeframe.

Ideally, documentation would be more evenly distributed throughout the project lifecycle.

13. From your perspective, what currently makes it difficult to achieve consistent documentation across projects?

The main difficulty lies in the educational context itself. Projects are short, student experience levels vary widely, and learning objectives are often prioritised over long-term continuity.

As a result, documentation quality depends heavily on guidance, templates, and institutional support structures rather than individual motivation alone.

14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?

It is important to be realistic about what student research can achieve. Not every project will produce outputs that are directly reusable.

However, by focusing on clear documentation, publication-oriented outputs, and stable project structures, we can gradually build a body of work that supports both education and longer-term research objectives.

Interview with Rea, Lab Supervisor

1. Can you describe your role in relation to the student research projects you have been involved in or supervised?

My involvement in student research projects is primarily content-driven and standards-oriented. As lead of the Critical Raw Materials track, I am responsible for ensuring that student work meets a minimum level of analytical rigor, data quality, and relevance to real-world supply chain challenges.

I do not see my role as primarily facilitative. Rather, I focus on assessing whether students are able to engage seriously with complex material, handle data responsibly, and produce work that can withstand scrutiny beyond the educational setting.

2. At the start of projects, what guidance are students typically given about how they are expected to document their research work? Can you give an example?

Students are informed that their research outputs must be defensible, data-driven, and clearly communicated. They are expected to keep track of their sources, datasets, and analytical steps to a degree that allows their conclusions to be evaluated.

In my experience, this guidance is sufficient. The problem is not a lack of instruction, but rather that students often underestimate the level of discipline required to carry out applied research properly.

3. Can you describe specific situations where documentation came up during project work with students? What happened in those moments?

Documentation issues typically arise when students are unable to justify their conclusions. When asked to explain how a particular insight was derived, it becomes apparent that underlying data is incomplete, poorly structured, or inconsistently referenced.

At that point, documentation is revealed not as a missing administrative step, but as a symptom of insufficient engagement with the data itself.

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to the original student team? Can you give an example?

Usability varies significantly, but in many cases it is limited. While reports may be written coherently, they often lack the data depth required for meaningful reuse.

In Critical Raw Materials projects, I frequently see descriptive narratives that are not supported by robust datasets. Without reliable data, documentation alone does not make the work reusable.

5. Which parts tend to be difficult for someone new to understand based only on what is written down or stored?

The main difficulty lies in the analytical gaps. It is often unclear how students moved from raw data to conclusions, not because the process was complex, but because it was never fully developed.

This results in work that appears finished on the surface but lacks internal coherence when examined closely.

6. Can you think of an example of an important decision or change during a project, and describe how and where it was documented, if at all?

A recurring issue is that projects change direction due to feasibility constraints, such as limited data availability or time pressure. These changes are usually not documented explicitly, but they are also rarely the result of deliberate methodological reflection.

Instead, they reflect reactive decision-making rather than structured research design.

7. If someone new wanted to understand how one important conclusion of a project was reached, what would they need to look at or read?

They would need access to the datasets used and a clear explanation of how those datasets were analysed. In many cases, this material is either missing or insufficiently developed.

Without solid data foundations, additional documentation does not meaningfully improve understanding.

8. During projects, is it clear who is responsible for documenting what, if anyone? How is that responsibility typically communicated or decided?

Responsibility is assumed to lie with the students. As emerging researchers, they are expected to take ownership of their work, including data handling and documentation.

When this responsibility is not taken seriously, it becomes visible in the quality of the final output.

9. What tools, templates, or systems are students expected to use to document work, and where is everything actually stored in practice?

Students have access to standard academic and collaborative tools. The specific choice of tool is less important than how it is used.

In my view, inconsistencies in storage or tooling are not the primary issue. The core problem is that students do not always use the tools they have with sufficient rigor.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, influences how motivated students are to document their work well?

Documentation serves to support data credibility and analytical transparency. It is not an end in itself.

Students are motivated to document their work properly when they understand that weak documentation reflects weak research. When documentation is treated as a formality, the underlying research quality is usually already compromised.

11. Based on your experience, what tends to get lost most often over time across projects?

What is lost most often is analytical depth. Data is collected, but not interrogated thoroughly. Assumptions remain implicit, and uncertainty is not addressed systematically.

This limits the long-term value of the work, regardless of how well it is written up.

12. How does documentation generally fit into project work from your perspective? Does it seem integrated into the work, or more like something added on top? Why?

Documentation is secondary to data work and analysis. When the research itself is weak, documentation cannot compensate for that.

From my perspective, improving research outcomes requires raising expectations around data competence, not expanding documentation requirements.

13. From your perspective, what currently makes it difficult to achieve consistent documentation across projects?

The main barrier is inconsistent student commitment and skill level. Applied research, particularly in domains such as Critical Raw Materials, demands a level of precision that many students are not yet prepared to deliver.

Without stronger selection, clearer performance expectations, or more rigorous assessment, inconsistency will persist.

14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?

If we want student research to contribute meaningfully to ongoing work, we need to be honest about its limitations. Not all student output is meant to be reused.

Continuity should be built on high-quality data and serious analytical effort. Without that, documentation becomes an attempt to preserve work that does not merit preservation.

Appendix B

VCH Lab Case Study Insight Analysis

A Diagnostic Assessment of Structural Barriers to Reproducible Research Capacity at the Value Chain Hackers Lab

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Document Type: Research Insight Analysis

Purpose

This document provides a structured diagnosis of barriers limiting reproducible research across five Value Chain Hackers Lab tracks. The analysis establishes the empirical foundation for the graduation project *Designing a Reproducible Research Framework for the VCH Lab.*

Abstract

This analysis presents a systematic evaluation of five applied research tracks conducted within the Value Chain Hackers Lab: Bakery, CRM, Organizational (Orga), Textile, and Cacao. Each project addresses real sustainability challenges and demonstrates strong student engagement. However, recurring weaknesses appear in documentation practices, methodological clarity, validation procedures, and the long term preservation of project records.

Taken together, these patterns indicate that the limitations are not isolated project issues but signs of a broader structural condition within the research environment. Reproducibility therefore emerges not as an individual student responsibility but as an institutional design requirement.

This analysis forms the empirical basis for the graduation project focused on developing a structured reproducible research framework and aligns with Windesheim's ambitions regarding research integrity, open science, and sustainable knowledge development.

The evidence shows that creativity alone is insufficient; reproducibility must be intentionally embedded within educational and operational structures.

Context and Analytical Lens

The Value Chain Hackers Lab provides students with an applied setting in which real world supply chain sustainability challenges are explored in collaboration with external stakeholders. While this model encourages initiative and problem solving, it also requires a level of research structure that ensures knowledge remains accessible and usable beyond a single project cycle.

For applied research to generate institutional value, projects must be understandable, traceable, and capable of being continued by future cohorts. This depends on clear documentation, transparent methodological choices, preserved data, and durable storage practices.

Each case was evaluated through the lens of reproducible research, with attention directed toward documentation integrity, methodological traceability, validation transparency, and the durability of stored project material. The selected tracks reflect variation in research maturity, storage behavior, and documentation practices within the lab, allowing patterns to be identified at a systemic rather than project specific level.

Three of the cases, CRM, Bakery, and Cacao, are supported by both project documentation and participant interviews, enabling deeper insight into how research practices unfolded in context. Textile and Orga are analyzed primarily through available project records and institutional materials. Their lack of interview data is analytically relevant in itself, as it reflects broader challenges related to documentation continuity and knowledge retention. An additional interview was conducted within the Beer track to capture experiential insight into documentation practices during an active project, strengthening the overall interpretation of research conditions within the lab, even though there was no documentation uploaded from that track to be analyzed.

Bakery Track

The Bakery track explored local sustainable sourcing practices through a stakeholder interview with a regional bakery chain. The only surviving project material is an audio recording without transcript, contextual notes, metadata, ethical documentation, or recorded analytical interpretation.

Although the interview represents genuine primary input, the absence of structured documentation prevents the insight from being verified, reconstructed, or meaningfully expanded. Knowledge generated through direct engagement therefore remains informational rather than research ready.

This case shows how quickly research value declines when foundational procedures such as transcription and contextual recording are not performed. Without these elements, even well executed fieldwork cannot contribute to cumulative institutional knowledge. The institutional Implication is that minimum documentation standards are necessary if applied projects are expected to strengthen reproducible research capacity.

CRM Track

The CRM track examined transparency in the lithium ion battery supply chain and produced substantial conceptual outputs, including digital traceability models. Students engaged actively with regulatory developments and reportedly conducted stakeholder conversations with organizations such as Umicore and TNO.

Despite this conceptual strength, the project lacks a coherent methodological record. Interviews were not transcribed, analytical steps were not documented, and multiple document versions exist without a traceable progression. As a result, the pathway from source material to final concepts remains difficult to follow.

This case highlights a critical distinction: innovative ideas alone do not establish research credibility. Reproducibility depends on demonstrating how insights emerged, how interpretations were formed, and how conclusions were justified.

Institutional implication:
Structured workflows, version control, and visible analytical procedures are required to anchor conceptual innovation in verifiable evidence.

Organizational (Orga) Track

The Orga track provides insight into the educational structure shaping the VCH Lab itself. The reviewed Start of Year framework emphasizes observation, sensing, ideation, and presentation. While effective for fostering creative exploration, it does not explicitly require research design clarity, validation cycles, ethical protocols, or standardized documentation practices.

Students are therefore primarily rewarded for producing outcomes rather than for demonstrating transparent research processes.

This case points to the institutional origin of many reproducibility challenges. When methodological discipline is not structurally embedded, expectations surrounding documentation remain implicit rather than operational.

Institutional implication:
Reproducibility cannot rely on individual initiative. It must be supported through educational design that establishes clear procedural expectations.

Textile Track

The Textile track investigated ethical apparel sourcing and sustainable production practices. Project work was conducted primarily within visual collaboration platforms such as Canva and Miro. While these tools supported ideation and mapping activities, the underlying research process was not preserved in a durable format.

No exported files, written analyses, or contextual explanations were retained within institutional systems. Once platform access expires, the project effectively disappears from organizational memory.

This represents the most severe level of reproducibility risk observed in the analysis: complete loss of retrievable project knowledge.

Institutional implication:
Allowing creative platforms to function as primary research environments introduces significant continuity risk. Archival requirements and export protocols are necessary safeguards for preserving institutional learning.

Cacao Track

The Cacao track produced *The Green Cacao Guide*, a visually cohesive synthesis of ethical sourcing practices and regulatory developments such as the EU Deforestation Regulation. The guide draws on credible secondary sources and communicates them effectively for an applied audience.

However, the methodological pathway behind the synthesis is not documented. Source selection criteria, analytical reasoning, and version history are not visible, and no formal validation process is recorded.

This case demonstrates that polished communication does not automatically equate to reproducible research. Without methodological transparency, even accurate summaries remain difficult to audit, extend, or critically evaluate.

Institutional implication:
Clear documentation of methodological choices is necessary to ensure that well presented outputs retain long term academic value.

Cross Case Synthesis

When examined together, the five tracks reveal a consistent lack of structural support for reproducible research within the VCH Lab. Similar weaknesses appear across projects regardless of topic, indicating that the issue is systemic rather than project specific.

Documentation is often incomplete, contextual information is limited, analytical pathways are not always visible, and storage practices are fragile. As a result, projects vary in the degree to which their knowledge can be retrieved or built upon. Some retain partial visibility yet lack methodological clarity, while others face near complete loss of institutional memory.

The unifying condition across the cases is the absence of operational structures that guide students in preserving transparent research processes. This supports the central premise of the graduation project: reproducibility must be intentionally designed into the research environment through structured workflows, documentation standards, validation checkpoints, and secure repositories.

Without these elements, applied projects may generate valuable insights but fail to contribute to cumulative institutional learning.

Institutional Relevance

This analysis provides Windesheim University with a structured diagnosis of barriers currently limiting the durability and credibility of applied research within the VCH Lab. The findings clarify how gaps in documentation, methodological visibility, and data preservation restrict long term knowledge development.

If unaddressed, these conditions risk producing research that is episodic rather than cumulative, reducing both academic defensibility and organizational learning capacity.

By identifying these structural issues across multiple cases, the analysis establishes a clear rationale for developing a reproducible research framework aligned with FAIR principles, ISO 27001 expectations, and national standards for practice oriented research.

Conclusion

The examined cases demonstrate that current reproducible research capacity within the VCH Lab is constrained primarily by structural rather than individual factors. Weaknesses in documentation, validation, and data stewardship limit the formation of a transparent and cumulative research tradition.

Embedding reproducibility into the operational design of the lab through clear templates, metadata practices, version control mechanisms, and validation routines will enable creative student work to evolve into a durable knowledge ecosystem capable of supporting sustained applied innovation.

Appendix C

Desk Research Sources and Relevance Mapping

Purpose

This appendix explains how desk research sources were selected, grouped, and applied within GP2. The purpose is to demonstrate methodological traceability by showing how each source category informed the empirical interpretation, theoretical grounding, and framework development. Sources were retained only if they directly contributed to explaining the diagnosed problem, strengthening methodological validity, or guiding solution design.

C1 Conceptual Foundations

This group of sources was used to interpret the diagnosed IS situation and explain why documentation breakdown occurs in applied research environments.

Work system alignment theory explains that outcomes emerge from the interaction of roles, tools, tasks, and evaluation criteria. This was used to interpret documentation weakness as structural misalignment rather than individual failure (Alter, 2013).

Socio technical systems theory supports the claim that stable behaviour requires alignment between technical structures and social expectations (Trist, 1981; Mumford, 2006). This informed the framework requirement that documentation must be embedded into workflow and supervision.

Knowledge conversion theory explains continuity loss as incomplete externalisation of tacit reasoning into explicit form (Nonaka and Takeuchi, 1995; Nonaka and von Krogh, 2009). This directly supports the requirement for visible decision logging.

Communities of practice theory explains how research routines become stabilised through shared norms and reinforcement (Wenger, 1998; Wenger et al., 2002). This informed the long term embedding logic of the framework.

Supply chain resilience research was used conceptually to illustrate that traceability breakdown increases systemic risk (Christopher and Peck, 2004). This strengthened the analogy between supply chain transparency and research transparency.

C2 Methodological Foundations

This group of sources supported research design, data collection, analytical sequencing, and framework derivation.

Design science research provides the methodological basis for developing and validating a structured artefact grounded in empirical diagnosis (Hevner et al., 2004; Peffers et al., 2007; Gregor and Hevner, 2013). These sources justify the framework as a research contribution rather than a recommendation.

Case study research principles support triangulation, chain of evidence, and structured analytical progression (Yin, 2018). These informed the alignment between document analysis, interviews, and synthesis.

Qualitative research methodology supports purposive sampling and interpretive validity (Patton, 2015).

Interview design literature informed the structured semi structured interview guide and question sequencing (Kallio et al., 2016; DeJonckheere and Vaughn, 2019).

Provenance and reproducible research literature defines reconstructability as visible analytical pathways linking sources to conclusions (W3C, 2013; Peng, 2011). These sources directly informed the transparency and verifiability criteria of the framework.

Design based research literature supports iterative refinement within educational environments (Anderson and Shattuck, 2012; Wang and Hannafin, 2005).

C3 Reproducibility and Governance Standards

This group provided external benchmarks used to evaluate the IS situation and derive framework criteria.

The FAIR principles define measurable standards for findability, accessibility, interoperability, and reuse (Wilkinson et al., 2016; Mons et al., 2017). These directly informed reuse and metadata requirements.

Open science literature positions documentation as a structural research responsibility (Nosek et al., 2015; Fecher and Friesike, 2014).

International governance frameworks emphasise structured data stewardship and documentation planning (OECD, 2021; Science Europe, 2021).

National research data management expectations further reinforce early planning and documentation embedding (Netherlands Organisation for Scientific Research, n.d.).

Institutional policies and infrastructure standards demonstrate that structured metadata and documentation are required for archival and reuse (SURF, n.d.; 4TU.ResearchData, n.d.; TU Delft Digital Competence Centre, n.d.; University of Twente, n.d.).

Lifecycle and stage awareness models reinforce that documentation must occur throughout the research process rather than only at final reporting (Gooch et al., 2019; Electronic Discovery Reference Model, n.d.).

Summary of Analytical Contribution

Conceptual sources were used to explain why documentation breakdown occurs.
Methodological sources were used to justify research design and artefact development.
Governance and reproducibility sources were used to benchmark the IS situation and define measurable framework criteria.

Together, these sources ensured that the transition from empirical diagnosis to structured framework specification remained theoretically grounded, externally aligned, and methodologically defensible.

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Appendix D

Desk Research Process Mapping for Theoretical Models

Introduction and Research Context

In GP1, I established a qualitative research design integrating desk research and field research within the Value Chain Hackers Lab at Windesheim University of Applied Sciences. The Lab operates as an applied research environment in which student teams collaborate with external organisations on challenges related to transparency, sustainability, and system resilience. Research unfolds under real world constraints, including fixed academic timelines, variation in student research experience, stakeholder availability, and operational complexity.

The study was grounded in the principle that applied research credibility depends on transparency, traceability, and reproducibility. When projects extend across student groups, weaknesses in documentation and continuity lead to repeated loss of research value. This creates micro level consequences for students, meso level consequences for supervisors, and macro level consequences when research practice diverges from broader expectations for open and reusable scholarship. These expectations are reflected in governance frameworks emphasising findability, accessibility, interoperability, reuse, and responsible stewardship of research outputs (Wilkinson et al., 2016; OECD, 2021; Science Europe, 2021).

Because the research was conducted in an uncontrolled applied setting, rigor was ensured through explicit methodological sequencing rather than experimental control. Following Yin (2018), I maintained a visible chain of reasoning linking empirical evidence, analytical interpretation, theoretical grounding, and framework derivation. Deviations from the original plan were implemented in a controlled and justified manner to preserve data quality.

The objective defined in GP1 was to design and validate a reproducible research framework capable of strengthening transparency, traceability, and reuse within student led applied research projects. This objective was aligned with design science principles, ensuring that the framework would respond to an observable problem, be grounded in validated knowledge, and be evaluated through explicit criteria (Hevner et al., 2004; Peffers et al., 2007).

I structured the study around a clear progression from the IS situation to the SOLL situation. The IS situation represents the empirically evidenced current state of research execution, documentation, and transfer within the Lab. The SOLL situation represents the justified target condition required to achieve higher standards of reproducibility and continuity. The analytical gap between these states forms the basis for deriving and validating framework requirements.

Identification of the IS Situation Through Field Research

I defined the IS situation empirically through document analysis and fully structured interviews. Document analysis examined completed project outputs to assess visibility of decisions, intermediate analytical steps, source usage, and validation logic. Structured interviews were conducted with former project leads and supervisory staff using purposive sampling to ensure alignment with the research question (Patton, 2015).

Triangulation was embedded by comparing stored project materials with participant accounts. Document analysis revealed what had been recorded, while interviews clarified how the research process unfolded and where documentation gaps occurred. This comparison strengthened interpretive credibility and reduced reliance on a single source (Yin, 2018).

Across projects, a recurring structural pattern emerged. Although outcomes often retained practical value, the analytical pathway from data to conclusion was not consistently reconstructable. Decision rationales and intermediate analyses were at times fragmented or insufficiently stored. Continuity frequently relied on verbal explanation rather than durable documentation. The issue was therefore interpreted structurally rather than individually.

The IS situation was defined as a systemic weakness in research traceability and continuity within a recurring applied laboratory environment.

Research Question Governance and Progressive Refinement

After defining the IS situation, the research remained governed by the research questions established in GP1.

The main research question defined the target: how to design and validate a reproducible research framework that makes student projects transparent, verifiable, and reusable in a consistent and institutionally reliable way. This question functioned as the structural compass of the study.

The original five GP1 subquestions structured the analytical progression:

The Define subquestion guided empirical diagnosis of structural and procedural barriers through document analysis and interviews.
The Design subquestions directed desk research toward external reproducibility standards and governance models capable of informing practical framework requirements.
The Execute subquestion introduced validation logic, requiring measurable criteria for assessing clarity, feasibility, and institutional alignment.
The Learn subquestion ensured that long term continuity and embedding were treated as core requirements rather than optional benefits.

During GP2, additional subquestions were introduced to increase analytical precision without altering the research direction. These included deeper examination of structural and incentive based breakdown, stage based documentation weakness, institutional benchmarking, and functional requirements for framework design. These additions were triggered by gaps identified during empirical analysis and served to refine, not redirect, the inquiry.

The progression therefore remained stable: diagnosis, explanation, specification, validation, and embedding. Analytical resolution increased while methodological control was preserved.

Institutional and System Level Benchmarking

To validate that the IS situation reflects a structural condition rather than a local anomaly, benchmarking was conducted across multiple levels.

At national level, funding requirements embed research data management planning at project inception (NWO, n.d.). At institutional level, universities align documentation practice with FAIR and defined stewardship responsibilities. At infrastructure level, repositories such as SURF Yoda and 4TU.ResearchData require structured metadata before archiving. At operational level, reproducibility verification services reinforce documentation as governed practice. At workflow level, lifecycle guidance such as The Turing Way emphasises embedding transparency across research stages.

Across levels, a consistent principle emerged: continuity and reuse are sustained when documentation is structured, embedded, and reinforced. This convergence strengthened the structural interpretation of the IS situation and provided an external benchmark for the SOLL condition.

Transition From GP1 to the Executed Research Process

GP1 anticipated surveys alongside interviews and document analysis. During execution, survey feasibility declined due to timing constraints. To preserve depth and contextual accuracy, fully structured interviews were prioritised. This adjustment maintained the original research intent while strengthening procedural insight (Yin, 2018).

The interview guide was deliberately constructed using fixed wording and sequencing to preserve comparability and reduce interpretive drift (Kallio et al., 2016; Patton, 2015).

Analytical Trigger for Desk Research

Field research functioned as the analytical trigger. Once the IS situation was defined as a structural weakness in traceability and continuity, desk research was initiated to identify validated theoretical models capable of explaining why such breakdown occurs and what structural conditions prevent recurrence.

Theory therefore served explanatory and operational purposes. It was not used to invent the problem, but to interpret and strengthen the empirically identified condition.

Research Strategy and Search Orientation

Desk research proceeded in three controlled stages.

Stage One established theoretical legitimacy by identifying models explaining traceability breakdown, workflow opacity, provenance weakness, and discontinuity in applied research environments.

Stage Two tested convergence by shifting the analytical lens toward educational design, organisational learning, and knowledge conversion. Independent searches revealing similar mechanisms increased confidence in structural stability.

Stage Three prioritised models directly supporting framework design, validation logic, and long term sustainability. Inclusion criteria required that retained theories explain breakdown, strengthen design legitimacy, guide operational structure, or define measurable evaluation standards (Hevner et al., 2004; Yin, 2018).

Throughout all stages, theoretical inclusion remained disciplined and traceable.

Stage One – Broad Theoretical Grounding

Stage One tested whether the diagnosed IS condition is recognised in established scholarship.

Work System Theory explained breakdown as system misalignment (Alter, 2013).
Design Science Research legitimised artifact development and structured evaluation (Hevner et al., 2004; Peffers et al., 2007).
Provenance theory clarified reconstructability requirements (W3C, 2013).
FAIR defined measurable reuse standards (Wilkinson et al., 2016).
Normalization Process Theory explained routine embedding dynamics (May & Finch, 2009).

Together, these models reinforced the structural interpretation of the IS situation.

Stage Two – Independent Convergence Testing

Stage Two introduced independent conceptual perspectives to test explanatory stability.

Design Based Research supported iterative improvement in educational settings.
Socio technical systems theory reinforced alignment between roles, incentives, tools, and routines (Trist, 1981; Mumford, 2006).
The SECI model explained continuity loss as incomplete knowledge externalisation (Nonaka & Takeuchi, 1995).
Open science lifecycle thinking emphasised stage embedded transparency (Nosek et al., 2015).
FAIR reappeared independently, strengthening its role as macro benchmark.

Convergence across independent domains confirmed that the IS situation reflects durable structural mechanisms rather than isolated interpretation.

Stage Three – Theoretical Prioritisation and Contextual Alignment

Stage Three narrowed the model set to those directly guiding framework development and evaluation.

Lifecycle logic was retained to ensure documentation across stages.
FAIR was retained to define measurable reuse criteria.
Design Science provided methodological structure for artifact validation.
Socio technical theory informed alignment requirements.
Knowledge conversion theory informed explicit externalisation of reasoning.
Communities of Practice theory supported long term embedding (Wenger, 1998).

This prioritised model set provides a focused and defensible foundation for operationalisation.

Stage Three Outcome

The IS situation reflects systemic misalignment, incomplete knowledge externalisation, weak lifecycle integration, and unclear reuse standards rather than individual negligence. The framework oriented response is academically legitimate under design science principles. Transparency and reuse can be evaluated through recognised standards such as FAIR. Continuity depends on embedding documentation within aligned socio technical structures and shared research practice.

Because each retained model serves a defined explanatory and operational role, the transition from IS diagnosis to SOLL framework specification remains structured, traceable, and methodologically defensible.

Appendix E

3.1 Answers to the Research Questions

Main Research Question

How can the Value Chain Hackers Lab design and validate a reproducible research framework that makes student projects transparent, verifiable, and reusable in a consistent and institutionally reliable way?

Section 2 demonstrated that current projects do not consistently achieve transparency, verifiability, or reuse. Documentation is often concentrated in final reporting. Decision rationales are not systematically logged. Intermediate reasoning steps are difficult to reconstruct. As a result, reuse across projects is limited.

Reproducibility scholarship shows that credible research depends on visible analytical pathways and preserved provenance rather than readable outcomes alone (Peng, 2011; Wilkinson et al., 2016). Governance frameworks similarly embed documentation as a structural requirement rather than optional practice (OECD, 2021).

The Lab must therefore implement a structured framework that embeds documentation throughout the research lifecycle, standardises file organisation and metadata, introduces mandatory decision logging and supervisor checkpoints, and aligns documentation quality with assessment criteria. Validation must be based on whether an independent reviewer can reconstruct reasoning pathways and whether documentation meets core reuse standards. This directly addresses the diagnosed structural weaknesses.

Subquestion 1 — Define

What structural and procedural barriers have limited reproducibility in past and current VCH projects?

Document analysis and structured interviews revealed recurring structural barriers. Documentation is not consistently embedded in workflow. Decision reasoning is not always logged. Links between sources, analysis, and conclusions are not always visible. File structures vary across projects. Documentation effort intensifies near submission rather than remaining stable during execution.

These findings indicate systemic misalignment rather than individual failure. Work System Theory explains that outcomes emerge from the interaction of roles, tools, incentives, and processes (Alter, 2013). Studies on student documentation confirm that incomplete provenance and inconsistent data management typically stem from structural design gaps rather than negligence (Chen et al., 2022). Reproducibility breakdown is therefore embedded in workflow and incentive structure.

Subquestion 2 — Design

Which reproducibility practices from other research environments can inform an approach suitable for the VCH Lab?

Benchmarking showed that mature research environments embed documentation early in project design, require structured metadata, define stewardship roles, and integrate review mechanisms. FAIR positions reuse as a core quality standard, while national governance bodies require formal data management planning from project inception (Wilkinson et al., 2016; OECD, 2021).

The transferable principle is clear. Reproducibility maturity emerges from governance structure, not individual discipline. Documentation becomes reliable when it is embedded, checked, and institutionally reinforced.

Subquestion 3 — Design

How can these principles be translated into a practical reproducible research framework for students and supervisors?

Translation requires converting governance logic into operational structure suitable for applied student research. Design Science Research requires solutions to be grounded in diagnosed problems and evaluated against explicit criteria (Hevner et al., 2004).

The framework must define lifecycle stages, introduce structured decision logs, standardise file naming and metadata, embed supervisor verification checkpoints, and align grading with documentation quality. Lifecycle models emphasise that documentation must span planning, execution, analysis, and reporting rather than appear only at final submission (Mons et al., 2017). The stage reference model supports distributing documentation duties across phases rather than concentrating them at the end.

The framework is therefore grounded both in empirical diagnosis and established research governance principles.

Subquestion 4 — Execute

How can the framework be tested for clarity, feasibility, and alignment with institutional policies?

Testing requires measurable evaluation logic. Validation must assess whether independent reviewers can reconstruct reasoning pathways, whether documentation completeness improves compared to prior projects, and whether practices align with recognised governance standards.

FAIR provides reuse and accessibility criteria. Governance frameworks provide compliance benchmarks (Wilkinson et al., 2016). Evaluation must therefore be structured and criteria based rather than subjective.

Subquestion 5 — Learn

How can the framework support long term knowledge continuity and educational value within the Lab?

Continuity requires documentation practices to become routine rather than temporary. Communities of Practice theory explains that stable practices emerge through shared norms and reinforcement (Wenger, 1998). Knowledge creation theory emphasises structured preservation and transfer for institutional learning (Nonaka and Takeuchi, 1995).

Because current documentation does not reliably support independent reconstruction or cumulative knowledge, long term value depends on structured repositories, defined responsibilities, and incentive alignment. Sustainability is therefore both structural and social.

Research Progression During GP2

During GP2, additional analytical questions deepened the diagnosis.

First, structural breakdown was traced to interacting elements including workflow design, unclear role ownership, limited validation checkpoints, and weak incentive alignment. This reframed documentation weakness as system interaction rather than isolated habit.

Second, stage based analysis clarified where breakdown occurs. Documentation appears strongest at final reporting and weaker during earlier execution stages. Lifecycle thinking enabled identification of stage specific vulnerability rather than treating documentation as binary.

Third, institutional benchmarking confirmed that the Lab condition reflects documentation maturity gaps rather than local anomaly. Mature environments embed documentation through planning requirements, metadata standards, and verification structures.

Fourth, framework design requirements were derived directly from this diagnosis. Mandatory decision logging, lifecycle structuring, supervisor checkpoints, metadata standards, and assessment alignment are not arbitrary features but structural corrections.

Fifth, long term embedding was reframed as organisational normalisation rather than template introduction. Sustainable change requires collective reinforcement and routine integration.

3.2 Synthesis and Strength of Conclusions

This section brings together all findings from Section 2 and explains clearly why the conclusions are justified. The central conclusion is straightforward. The Value Chain Hackers Lab does not yet operate with a structured system that consistently guarantees transparency, verifiability, and reuse across student research projects. Final reports are usually completed and readable, but the full pathway from sources to analysis to conclusions is not always documented in a way that allows independent reconstruction.

This conclusion is not based on a single observation. It emerges from the combined evidence of project file reviews, structured interviews, and institutional benchmarking.

Across five completed projects, the review of stored materials showed that final deliverables are present, but intermediate reasoning steps are not always clearly visible. Analytical decisions can often be inferred, yet they are not consistently recorded as explicit, traceable steps. This means that a new student or supervisor can see what was concluded, but may struggle to independently verify how those conclusions were reached.

The interviews reinforce this pattern. One student explained that “You see the result, but not everything that led up to it.” Another described how important shifts in direction “just appeared in the work later,” without being formally recorded as decisions. These statements show that reasoning took place, but that it was not consistently captured. The issue is therefore not a lack of thinking, but a lack of structured recording.

The lab leads describe the same structural condition from a supervisory viewpoint. Chris stated that documentation “takes second place” compared to presentation and visible performance in final evaluations. Rea explained that when documentation becomes a formality, research quality is already affected. Maxime observed that documentation “often becomes more prominent toward the end of projects,” which aligns with students’ accounts of stronger effort close to submission deadlines rather than throughout the full project. When both students and supervisors describe the same pattern, the conclusion gains strength.

Why This Pattern Occurs

The findings can be clearly explained using established theory.

Work System Theory shows that behaviour in an organisation is shaped by how tasks, roles, tools, and evaluation criteria are aligned (Alter, 2013). If grading and public evaluation focus mainly on final presentation, students will logically prioritise those elements. Documentation will not become stable practice unless it is visibly required and assessed.

Socio technical systems thinking explains that durable behaviour requires both technical structure and social reinforcement (Trist, 1981; Mumford, 2006). Templates, checkpoints, and defined responsibilities must support expectations. Without such structure, documentation depends on individual initiative rather than institutional routine.

Research on student documentation practices supports this interpretation. Studies show that when expectations are unclear, documentation quality varies significantly (Chen et al., 2022). Research on barriers to rigorous practice demonstrates that breakdowns are often driven by structural and incentive conditions rather than by individual unwillingness (Lalu et al., 2023). The Lab findings reflect this same mechanism.

Reproducibility standards further clarify the gap. The FAIR principles define research quality in terms of findability, accessibility, interoperability, and reuse (Wilkinson et al., 2016). Reuse requires that another person can follow the reasoning from source to conclusion. Peng (2011) similarly argues that reproducible research demands transparent reporting of analytical steps, not only final outcomes. In the reviewed Lab projects, final results are available, but the reasoning chain is not always independently verifiable.

The Electronic Discovery Reference Model strengthens this explanation by clarifying where weaknesses occur within the research process. The model distinguishes stages such as identifying information, processing it, reviewing decisions, and producing outputs (EDRM, n.d.). When this lens is applied to the Lab projects, documentation appears strongest at the final reporting stage and weaker during earlier working stages. This suggests that documentation is often completed retrospectively rather than maintained consistently during the project.

Institutional benchmarking confirms that this issue is structural rather than unique. National funders such as the Netherlands Organisation for Scientific Research require early research data management planning (NWO, n.d.). International governance bodies emphasise structured documentation and reuse as core research responsibilities (OECD, 2021; Science Europe, 2021). Universities formalise these expectations through research data management policies aligned with FAIR principles (University of Twente, n.d.). Infrastructure providers such as SURF and 4TU.ResearchData require structured metadata before research can be archived and reused (SURF, n.d.; 4TU.ResearchData, n.d.). These systems demonstrate that documentation maturity emerges from embedded governance, not individual effort alone.

Compared to these standards, the Lab currently relies more on student discretion than on enforced structure. Supply chain research shows that when traceability breaks down at one stage, risk spreads and resilience decreases (Christopher and Peck, 2004). In the Lab context, weak documentation limits continuity and reduces cumulative learning.

The strength of these conclusions lies in convergence. Project reviews, student interviews, lab lead interviews, theory, and institutional benchmarks all point toward the same structural explanation. No significant contradictions were found. Differences relate to emphasis, not to the existence of the problem itself.

These conclusions are also appropriately bounded. The study focuses on one applied research lab and does not claim universal generalisation. However, because the findings align with recognised reproducibility and governance standards (Wilkinson et al., 2016; OECD, 2021; Nosek et al., 2015), the conclusions are transferable to similar student led research environments.

In summary, the synthesis shows that the core issue is structural misalignment. Documentation is not yet embedded into research workflow as a guided, visible, and assessed component. Without structural embedding, transparency remains partial, verifiability remains limited, and reuse remains unreliable. This logically leads to the framework specifications developed in Section 3.3, where solution requirements are defined directly in response to the weaknesses identified here.

3.3 Specifications and Criteria for Solution Development

This section translates the conclusions from Sections 2 and 3.2 into a practical solution that can be implemented and enforced. The solution is a reproducible research methodology framework that is integrated as a Lab level university policy. The policy approach is necessary because the findings show that documentation quality is mainly shaped by what is expected, what is checked, and what is rewarded. When documentation is treated as optional or secondary, students logically prioritise what is most visible during assessment. A policy integrated framework corrects this by making documentation a required part of the research process, with clear responsibilities, clear checkpoints, and clear quality criteria (Alter, 2013; Trist, 1981; Mumford, 2006; Lalu et al., 2023).

The framework is designed as a step by step structure that runs from project start to final archiving. It is grounded in design science logic, meaning the solution is designed to address a real problem using clear requirements and clear evaluation criteria, rather than vague advice (Hevner et al., 2004; Peffers et al., 2007; Gregor and Hevner, 2013). It is also aligned with reproducibility and research governance standards that treat documentation as a core research responsibility, including FAIR and broader research data management expectations (Wilkinson et al., 2016; OECD, 2021; Science Europe, 2021). A lifecycle lens is used to ensure documentation happens throughout the work and not only at the end. For this lifecycle structuring role, the Electronic Discovery Reference Model is used only as a stage reference to check whether information is being captured during source identification, processing, review, and final production, instead of being left for late reporting (EDRM, n.d.). Provenance thinking also supports this requirement by emphasising that the history of decisions and sources must remain visible if work is to be checked and reused (W3C, 2013; Peng, 2011).

Policy intent, scope, and enforcement logic

The policy intent is to make every student project transparent, verifiable, and reusable in a consistent way. Transparency means the research pathway is visible, including what sources were used and why key decisions were made. Verifiability means another person can follow the recorded steps and check whether conclusions match the documented sources and analysis. Reuse means a new student team can continue the project using stored materials without needing private explanations. Consistency means these practices are required for every project, not left to personal preference (Wilkinson et al., 2016; Peng, 2011; OECD, 2021).

The policy scope applies to all student projects conducted under the Lab that involve research and external or internal stakeholders. Enforcement is achieved through three mechanisms. First, every project starts with the same required structure and templates. Second, supervisors check specific documentation checkpoints during the project, not only at the end. Third, the final submission is not considered complete unless the required documentation package is included and meets the minimum criteria for transparency, verifiability, and reuse (Nosek et al., 2015; Science Europe, 2021).

Step 1: Project onboarding and documentation setup

What happens at this step is that the project is set up as a documented research workspace before data collection begins. This includes a standard folder structure, standard file naming rules, and a clear place where decisions and sources will be recorded. The policy requires this at the start because the findings showed that when documentation is delayed, the reasoning chain becomes harder to reconstruct later. This step directly supports reuse and reduces loss of context (Wilkinson et al., 2016; Gooch et al., 2019; Mons et al., 2017).

How it is done is by using a standard project template that includes a decision log, a source log, and a clear structure for working files and final outputs. The reason this is needed is that research data lifecycle standards consistently show that documentation quality improves when structure is introduced early and followed consistently (Gooch et al., 2019; Science Europe, 2021). It also aligns with institutional approaches where documentation is treated as an early planning requirement rather than an end stage task, such as the expectation that research data management is planned from the beginning of a project (NWO, n.d.; OECD, 2021).

Who is responsible is the student project lead, with the supervisor verifying that the workspace meets the minimum setup standard before the project proceeds. The supervisor role is necessary because studies on research practice show that structure becomes stable only when it is reinforced through oversight and shared routine rather than left to individual discipline (May and Finch, 2009; Wenger, 1998).

Step 2: Research execution with continuous documentation

What happens in this step is that documentation is maintained during normal research work, not only during final writing. The policy requires that students record key decisions, changes in direction, and how sources are used. This is required because the analysis showed that decision rationales and intermediate reasoning steps are often missing or hard to locate in stored materials, which weakens verifiability and reuse. Continuous recording directly addresses that weakness (Peng, 2011; Wilkinson et al., 2016).

How it is done is by requiring short, simple updates in a decision log and source log during the project. This does not need to be long or complex. The goal is to make the reasoning visible. This aligns with reproducible research guidance that stresses that the steps behind results must be recorded, not only the results themselves (Peng, 2011; Nosek et al., 2015). It also fits lifecycle based models that treat documentation as part of ongoing work rather than a final reporting activity (Gooch et al., 2019; Mons et al., 2017).

Who is responsible is the full student team, but the project lead owns the completeness of the logs. The supervisor checks progress during scheduled review moments. This shared ownership model is consistent with the idea that stable research practice is built through shared routines and role clarity (Wenger, 1998; Wenger et al., 2002).

Step 3: Supervisor checkpoints that verify traceability before final delivery

What happens in this step is that supervisors carry out structured checks during the project, not only after final submission. This step exists because the findings showed that documentation effort rises near final delivery and is shaped by what is evaluated. If checks only happen at the end, students will logically treat documentation as a final stage activity. Checkpoints shift this by making documentation quality visible earlier (Alter, 2013; Lalu et al., 2023).

How it is done is by using a short checklist that verifies whether decisions are recorded, sources are linked to claims, and working steps are understandable. This directly supports the requirement that another person can follow the reasoning chain without relying on personal explanation, which is central to verifiability and reuse (Peng, 2011; Wilkinson et al., 2016). Institutional benchmarking also supports this logic because verification practices are increasingly treated as part of responsible research support, not a personal preference (TU Delft Digital Competence Centre, n.d.).

Who is responsible is the supervisor, and the student team is required to correct gaps immediately after each check. This reflects good governance practice where research quality is reinforced through routine review rather than retrospective correction (Science Europe, 2021; OECD, 2021).

Step 4: Final submission requirements that make documentation part of the deliverable

What happens here is that the final project submission includes both the report and a required documentation package. This is necessary because the findings show that documentation is currently not treated as an integral part of what is valued in final evaluation moments. The policy changes this by making documentation an assessed requirement, not an optional add on (Alter, 2013; Nosek et al., 2015).

How it is done is by requiring a clear bundle that includes the final report, the decision log, the source log, and the stored research materials organised according to the standard structure. This aligns with governance frameworks that treat documentation and stewardship as part of responsible research practice (OECD, 2021; Science Europe, 2021). It also aligns with the FAIR logic that reuse requires clear structure and context, not only a final narrative (Wilkinson et al., 2016; Mons et al., 2017).

Who is responsible is the student team, with the supervisor confirming completeness before submission is accepted as final.

Step 5: Archiving and reuse readiness

What happens at this step is that the project is prepared for storage and future reuse through clear metadata and packaging rules. This step exists because reuse depends on future findability and interpretability. Benchmarking shows that mature research environments require structure before archiving, including metadata that explains what the materials are and how they connect (SURF, n.d.; 4TU.ResearchData, n.d.). This is directly relevant to the Lab because the key risk is not missing outputs but missing context that allows someone else to continue the work.

How it is done is by requiring short metadata fields that describe the project aim, methods used, key datasets or sources, and where decision logs and final conclusions connect. This approach matches research data lifecycle thinking where documentation supports long term stewardship and reuse (Gooch et al., 2019; Mons et al., 2017). It also aligns with the principle that documentation should be integrated into the workspace and not added only at the end, as reinforced by reproducibility guidance such as The Turing Way (The Alan Turing Institute, n.d.).

Who is responsible is the project lead, with a final check by the supervisor or Lab staff responsible for storage quality.

Framework criteria that define success

The framework is successful only if it meets the four outcome criteria in a way that can be checked.

Transparency is achieved when a reader can see which sources informed key claims and can see why major decisions were made, supported by recorded decision and source logs (Peng, 2011; Wilkinson et al., 2016). Verifiability is achieved when another student or supervisor can follow the recorded steps and judge whether conclusions match the documented sources and analysis, without needing private explanation (Peng, 2011; Nosek et al., 2015). Reuse is achieved when a new student team can continue the work using the archived package, because the structure and metadata make the materials understandable and connected (Wilkinson et al., 2016; Mons et al., 2017; SURF, n.d.). Institutional consistency is achieved when the same required structure, checkpoints, and submission rules apply to every project so that documentation quality is not dependent on personal habits (Science Europe, 2021; OECD, 2021; Alter, 2013).

By structuring the solution as a policy integrated framework with clear steps, clear responsibilities, and clear criteria, the Lab gains a practical mechanism that directly addresses the weaknesses identified in Section 2 and supports the conclusions in Section 3.2. The result is a framework that is not only conceptually correct, but also enforceable, checkable, and suitable for stable use inside a university research lab setting.

3.4 Validity, Certainty, and Remaining Unknowns

This section evaluates how strong the conclusions are, how certain they can be considered within the defined scope, and what limitations or open questions remain. The goal is to demonstrate that the conclusions and the proposed framework are logically derived, theoretically grounded, and externally benchmarked, while also acknowledging clear boundaries.

Validity of the Conclusions

The conclusions are supported by three converging foundations: structured case analysis, structured interviews, and staged desk research aligned with established research governance and reproducibility standards.

First, five completed projects were analysed using the same structured lens. Across different domains, including municipal resource coordination, food supply chains, textile sustainability, cacao transparency, and internal research structuring, the same weaknesses appeared: limited visibility of reasoning steps, weak decision logging, and inconsistent storage logic. The repetition of these patterns across different contexts increases confidence that the issue is structural rather than project specific (Yin, 2018).

Second, seven structured interviews with student leads and Lab leadership independently reinforced these findings. Students described unclear documentation expectations and grading driven prioritisation. Lab leaders confirmed that documentation is often secondary to presentation and that evaluation signals influence behaviour. The consistency across execution and supervisory roles strengthens internal validity because the same problem is described from multiple positions (Patton, 2015).

Third, the findings align with established research on documentation breakdown and incentive driven research behaviour (Chen et al., 2022; Lalu et al., 2023). Reproducibility scholarship consistently emphasises visible reasoning pathways, provenance clarity, and lifecycle documentation as necessary for credible and reusable research (Peng, 2011; Wilkinson et al., 2016). This alignment strengthens external validity by showing that the diagnosed weaknesses reflect recognised research system challenges rather than isolated student behaviour.

Strength of the Framework Derivation

The proposed framework is not an intuitive proposal. It is logically derived from the diagnosed breakdowns and systematically benchmarked against multiple levels of external standards.

At theoretical level, the design follows structured solution logic consistent with design science principles, where a real organisational problem is addressed through clearly defined requirements and evaluation criteria (Hevner et al., 2004; Peffers et al., 2007).

At governance level, the framework aligns with FAIR principles, OECD data sharing recommendations, and Science Europe guidance that position documentation as a structured and enforceable research responsibility (Wilkinson et al., 2016; OECD, 2021; Science Europe, 2021).

At institutional and infrastructure level, benchmarking included national funding requirements, structured research data management policies, repository standards, and reproducibility verification practices. These demonstrate that mature research environments embed documentation structurally through planning requirements, metadata standards, and verification services rather than leaving it to individual preference (NWO, n.d.; University of Twente, n.d.; SURF, n.d.; 4TU.ResearchData, n.d.; TU Delft Digital Competence Centre, n.d.).

At workflow level, lifecycle integration guidance such as The Turing Way demonstrates that documentation must be embedded into daily research activity, not concentrated at final reporting (The Alan Turing Institute, n.d.).

The use of lifecycle sequencing logic, supported by FAIR lifecycle thinking and structured stage awareness including the Electronic Discovery Reference Model, ensured that documentation expectations were defined across project stages rather than treated as end stage reporting (Wilkinson et al., 2016; Mons et al., 2017; EDRM, n.d.).

Because the framework was derived through this layered process, from local diagnosis to theoretical validation to institutional benchmarking, its requirements are methodologically defensible and externally justified.

Certainty Within Scope

The certainty of the conclusions is strong within the defined institutional scope. The study does not claim universal generalisation. It provides analytically grounded conclusions about one applied research lab operating within a Dutch university context.

The convergence across case analysis, structured interviews, and external benchmarking reduces the likelihood that the findings are accidental or based on isolated experience. The use of predefined constructs and structured instruments further reduces interpretation drift (Kallio et al., 2016).

The framework therefore rests on converging evidence rather than assumption.

Remaining Unknowns

Despite this strength, some boundaries remain.

First, while the framework is theoretically justified and benchmarked against comparable research environments, long term behavioural adoption must still be observed in practice.

Embedding stable routines depends on shared meaning, supervision consistency, and institutional reinforcement over time (Wenger, 1998; May and Finch, 2009).

Second, although the framework aligns with governance and data stewardship standards, its operational integration into all university systems, including formal policy codification and digital infrastructure alignment, requires institutional coordination beyond the scope of this research.

Third, incentive structures were identified as a central factor shaping documentation behaviour. While the framework addresses this by integrating documentation into evaluation logic, further observation will be required to assess how strongly behaviour changes once enforcement mechanisms are active.

Overall Assessment

Within the defined scope, the conclusions are strongly supported by structured case evidence, structured interviews, theoretical grounding, lifecycle structuring logic, and multi level institutional benchmarking.

The diagnosis of structural documentation weakness is highly certain. The framework design is logically derived and externally validated.

Uncertainty remains primarily at the level of long term implementation and behavioural embedding, not at the level of structural analysis or conceptual justification.